**Vishwakarma Awards 2026–27**

*Project Progress Update*

**Title:** Clear-Flow: Smart Manhole Monitoring and Overflow Prediction System
**Theme:** Sustainable Cities
**Sub Theme:** Clean Water, Sanitation & Air Quality Monitoring
**Project Status:** Lab Prototype / Field Experiment

## 1. Original Prototype and What We Are Adding

Our original prototype is very basic. It's just a normal ESP32 board connected to an ultrasonic sensor (AJ-SR04M), sitting on a small benchtop rig we built using two water containers to simulate a manhole filling up. The ESP32 reads the water level and sends it to a Blynk app over WiFi, mainly to check if the readings made sense. That's it — no LoRa, no custom PCB, no waterproof enclosure, no AI, and nothing running on a cloud dashboard. It was really just a way to prove the ultrasonic sensor could measure rising water accurately.

Since applying, we are working on adding LoRaWAN in place of WiFi, since a real manhole will not have WiFi, and a device that needs constant internet does not really work underground. We are also starting to work out the sleep/wake logic — right now the ESP32 sends data constantly, which we know would drain a battery in days, not years. On the dashboard side, we are still at the design stage for the weather-check logic that tells a blockage apart from rain, and the AI model that would predict cleaning schedules. None of this is built yet — it is all work in progress.

## 2. How This Update Improves the Original Work

The WiFi and Blynk version is fine to prove the sensor works, but it is not usable in a real manhole. Most underground manholes will not get a WiFi signal, and running WiFi constantly needs a lot of power, so the battery would not last more than a few days. Once we move to LoRaWAN, the device will not need any SIM card, WiFi, or mobile data at the node at all, so it can realistically sit underground for years on a single battery, which is much closer to what a city would actually need.

Keeping the weather check on the dashboard side, instead of trying to do it on the device, also keeps the node itself simple and cheap, while the smarter part — telling rain apart from a real blockage, and predicting a cleaning schedule — happens on the cloud side, where there is actual computing power to do it properly. Compared to industrial radar or ultrasonic sensors that cost lakhs of rupees per unit, our version uses cheap, easily available parts, so it can realistically be placed on many manholes instead of just one or two expensive ones.

## 3. Biggest Insight, Hurdle, and Next Steps

The biggest thing we learned from testing was how badly the raw sensor readings jump around from splashing. Even small waves in our test rig caused the distance reading to swing a lot, which would have triggered a false alert if we had not smoothed it out in code. We did not expect the ultrasonic sensor to be this sensitive to surface disturbance. Testing also showed us that continuously sending readings over WiFi drained the ESP32's power fast, which is what pushed us to plan the move to LoRa along with a proper sleep-wake cycle instead of always-on monitoring.

Going forward, here is the order we plan to integrate things in:

1. **LoRaWAN module with the ESP32** — replacing the current WiFi connection.
2. **Sleep and wake logic** — deep sleep most of the time, waking up more often only when the water level is rising.
3. **LoRa gateway** — to receive the node's signal and pass it on to the internet.
4. **Cloud dashboard** — in place of the current Blynk setup, fed through the gateway.
5. **Blockage alert logic** — flagging a manhole once its water level keeps climbing.
6. **Weather API integration** — to tell a flash flood apart from an actual blockage.
7. **Custom PCB** — replacing the current breadboard wiring.
8. **Waterproof enclosure** — sealing the finished electronics for real underground use.
9. **Full integration test** — checking the complete system together before any real-world placement.